

ELK ISLAND NATL PARK, AB, Canada

WMO: 712380

Lat: 53.6825N

Lon: 112.8681W

Elev: 2350

StdP: 13.49

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.1	-19.8	-33.2	0.9	-25.9	-26.6	1.4	-19.4	15.0	12.7	13.3	10.7	1.1	130	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	82.8	64.7	79.7	63.0	76.7	61.4	67.6	78.4	65.3	76.2	63.3	73.6	4.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	95.9	74.1	61.1	87.8	70.7	59.1	81.6	68.2	33.4	78.5	31.5	76.3	29.9	73.8	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	11.9	10.1	-35.3	88.7	7.4	3.9	-40.6	91.5	-44.9	93.8	-49.1	96.0	-54.5	98.8		
(5)			-35.3	71.8	7.3	2.7	-40.5	73.7	-44.8	75.3	-48.8	76.8	-54.1	78.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	11.5	14.6	24.1	38.8	51.2	58.3	62.6	60.7	51.8	39.7	24.6	13.4		
	DBStd	21.51	16.28	14.46	14.09	9.50	8.25	5.76	5.46	6.25	8.04	8.52	12.14	14.65		
	HDD50	5731	1192	990	803	348	85	6	1	4	73	331	763	1135		
	HDD65	10032	1657	1410	1268	785	430	213	112	161	399	784	1213	1600		
	CDD50	1254	0	0	0	13	123	254	392	334	126	12	0	0		
	CDD65	84	0	0	0	0	4	11	39	27	3	0	0	0		
	CDH74	1274	0	0	0	2	93	178	510	375	113	3	0	0		
	CDH80	248	0	0	0	0	12	26	115	74	21	0	0	0		
	WSAvg	3.9	3.5	3.6	4.5	5.2	4.8	3.8	3.4	3.1	3.4	4.3	3.8	3.4		
	PrecAvg	18.3	0.9	0.5	0.9	1.3	2.0	3.1	3.7	2.0	1.5	0.9	0.7	0.8		
	PrecMax	25.4	2.7	1.1	2.2	2.7	4.9	6.0	7.4	4.4	3.2	2.2	1.7	2.0		
(15) Precipitation	PrecMin	10.8	0.2	0.1	0.2	0.0	0.3	1.0	1.7	0.6	0.5	0.1	0.1	0.2		
	PrecStd	3.0	0.7	0.3	0.5	0.8	1.0	1.4	1.3	0.9	0.8	0.6	0.5	0.5		
	DB	47.8	49.9	56.5	71.2	82.1	83.2	88.0	86.4	83.1	71.3	54.6	48.1			
	MCWB	38.4	38.0	43.1	50.2	57.4	62.3	67.6	68.2	62.2	54.3	43.5	37.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	42.3	44.1	50.9	65.2	76.5	79.1	83.0	81.8	77.1	62.7	47.5	42.0			
	2% MCWB	35.5	35.5	39.8	47.6	55.8	60.6	66.5	65.5	59.2	49.3	39.3	35.1			
	5% DB	38.4	39.4	46.7	59.5	72.0	75.3	79.6	78.2	72.1	57.5	43.1	37.9			
	5% MCWB	32.9	33.0	37.6	44.6	53.5	59.2	64.6	63.9	57.5	46.1	36.6	32.1			
	10% DB	34.0	34.9	42.4	54.8	67.5	71.8	76.2	74.7	66.6	53.0	39.2	33.5			
	10% MCWB	30.1	30.4	35.3	42.3	51.2	57.5	62.9	62.0	54.6	43.9	34.2	29.5			
	0.4% WB	38.8	39.6	43.7	51.3	61.6	65.9	71.4	71.3	64.2	54.5	44.0	38.7			
	0.4% MCDB	46.2	48.5	54.9	68.5	73.8	77.3	82.7	83.1	78.1	68.7	53.9	46.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.0	36.1	40.4	48.3	57.7	63.3	68.3	67.6	60.9	50.5	40.1	35.3			
	2% MCDB	41.8	43.0	49.4	63.4	72.2	74.6	79.4	77.9	73.9	60.1	46.3	41.6			
	5% WB	33.3	33.3	38.1	45.6	55.1	61.1	65.9	65.1	58.2	47.5	37.2	32.5			
	5% MCDB	38.1	39.0	45.9	57.8	68.8	71.8	77.1	75.7	70.2	55.6	42.2	37.3			
	10% WB	30.0	30.4	35.6	43.1	52.7	59.2	63.9	62.8	55.6	44.8	34.4	29.5			
	10% MCDB	34.0	35.2	42.0	53.1	65.3	69.7	74.0	72.8	65.5	51.8	38.7	33.6			
	MDBR	20.4	23.7	22.7	22.0	22.9	21.6	22.2	22.7	22.5	19.1	16.8	19.0			
	5% DB	23.3	29.4	27.1	29.2	29.5	27.6	27.6	28.2	31.7	27.6	21.1	21.5			
(36) Mean Daily Temperature Range	MCWBR	17.1	21.6	17.3	15.1	13.4	13.0	13.5	14.2	15.8	15.0	13.7	15.7			
	MCDBR	22.6	28.2	25.5	27.3	26.8	23.8	25.2	25.9	29.5	25.8	19.6	20.6			
	MCWBR	17.0	20.9	16.6	14.9	13.1	12.5	13.6	14.0	15.3	14.8	13.2	15.6			
	MCDBR	22.6	28.2	25.5	27.3	26.8	23.8	25.2	25.9	29.5	25.8	19.6	20.6			
(40) Clear-Sky Solar Irradiance	taub	0.254	0.258	0.309	0.326	0.339	0.353	0.353	0.361	0.305	0.277	0.251	0.242			
	taud	2.307	2.365	2.304	2.383	2.396	2.375	2.416	2.349	2.501	2.551	2.449	2.281			
	Ebn at Noon	233	274	278	288	288	284	282	272	279	266	238	213			
	Edh at Noon	18	23	31	33	35	36	34	34	26	20	16	15			
(44) All-Sky Solar Radiation	RadAvg	304	633	1122	1482	1816	1908	1949	1603	1106	634	302	223			
	RadStd	34	49	86	138	135	124	96	111	114	76	34	21			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.01	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page